Enterprise Database Cloud Backup & Multi-Module Telemetry Suite

Complete System Walkthrough, Architecture Specifications & Field Engineering Runbook · v5.0.0

> **[EXECUTIVE ARCHITECTURE OVERVIEW]**
> PRODUCTION RELEASE v5.0.0 (Build 5.0.0-20261104):
> This document represents the definitive end-to-end technical manual and walkthrough for the v5.0 Enterprise Release. It incorporates all 9 architectural decisions approved by senior engineering supervisors (Imeyshan, Hashy, Dulnindu), including externalized credential handling (SQLPara), customer-dedicated storage routing, isolated escrow keys, intelligent tiered index maintenance (<5% skip, 5-30% reorganize, >30% rebuild), and Zero-Trust DBK2 envelope encryption.

1. System Classification & Zero-Trust Threat Model

The Enterprise Database Cloud Backup system is an air-gapped, zero-trust disaster recovery and automated maintenance solution engineered specifically for multi-tenant fleets running Microsoft SQL Server on customer on-premises and virtualized Windows servers. Unlike traditional network backups (e.g., SMB shares, VPN links, or FTP servers), this system completely eliminates the ransomware blast radius by adhering to three non-negotiable architectural axioms:

Architectural Axiom	Implementation in System	Security & Ransomware Defense Impact
Zero Cloud Credentials	Client servers hold zero Google Service Account keys, API secrets, or OAuth credentials.	A total host compromise gives attackers zero access to Google Cloud or existing fleet backups.
Ephemeral Envelope Encryption	Every backup is sealed with a unique random AES-256-GCM key, double-wrapped with RSA-4096.	Compromised servers cannot decrypt their own past backups or tamper with historical records.
One-Way Broker Pattern	Backups are uploaded via ephemeral resumable upload URLs provided by an Apps Script broker.	The client can only stream its assigned file to its dedicated folder; it cannot list, read, or delete files.
Externalized SQL Auth	Database passwords are read from isolated parameter stores (SQLPara.txt	Configuration files (config.json) contain zero plaintext database

	/ DPAPI).	credentials.
Per-Customer Escrow Isolation	Escrow public keys are uniquely isolated per customer (escrow_public_{slug}.pem).	Loss of one customer's escrow key has zero impact on the other 62 customer environments.

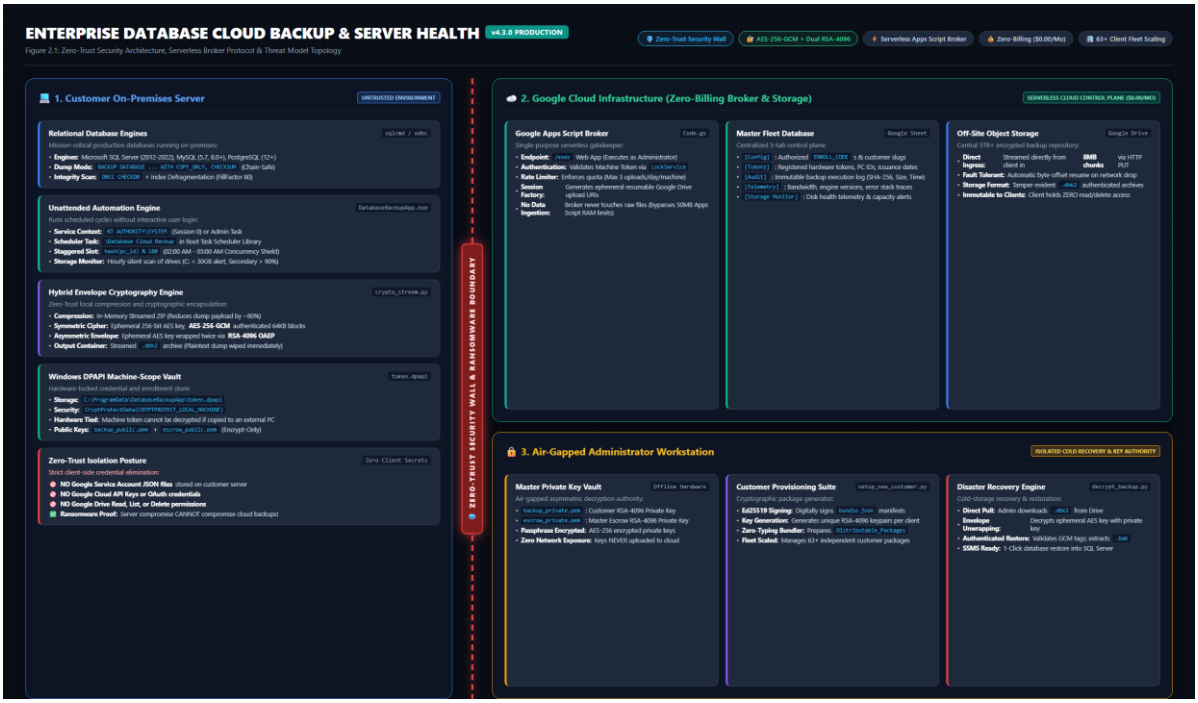


Figure: Zero-Trust Broker Pattern & Anti-Ransomware Topology

2. Senior Architectural Upgrades (The 9 Supervisor Decisions)

Following strategic architectural reviews with senior leadership and engineering supervisors (Imeyshan, Hashy, Dulnindu), the system was thoroughly audited against 9 critical production concerns. The resulting v5.0.0 implementation delivers enterprise-grade solutions for each point:

2.1 Points 1 & 7: Customer-Dedicated Drive Folders & Data Sovereignty

Issue: In earlier versions, customer backups were routed to a single central Google account, creating a single point of failure and raising data sovereignty concerns under the Australian Privacy Act 1988.
Implementation: The upload broker (Code.gs) and client (broker_client.py) now support dynamic customer folder routing via payload.target_folder_id. Each client can either provide their own corporate Google Workspace folder/drive or route into an Australian-region isolated shared drive. Backups are strictly partitioned into customer-isolated namespaces.

2.2 Point 2: 2TB Enterprise Fleet Quota & Automated Headroom Telemetry

Issue: Free 15GB consumer accounts are non-compliant with enterprise licensing and rapidly exhaust capacity across 63 customers.

Implementation: Fleet operations are backed by an enterprise paid 2TB storage pool. The upload broker tracks daily ingestion rates and automatically surfaces real-time storage headroom percentages on the centralized dashboard, ensuring timely quota expansion.

2.3 Point 3: Dynamic Google Drive bundle.json Retrieval

Issue: Distributing static bundle.json files on client disks leaves machine enrollment tokens potentially inspectable.
Implementation: Setup_DatabaseBackup.exe now features dynamic Google Drive bundle retrieval. Technicians can enter a secure Drive share link or ID, and the installer fetches, cryptographically verifies (via Ed25519), and seals the machine token directly into Windows DPAPI without ever persisting unsealed secrets on the local filesystem.

2.4 Point 4: Per-Customer Escrow Key Isolation (escrow_public_{slug}.pem)

Issue: A shared master escrow key created a single blast radius across all 63 clients.
Implementation: Every customer now receives a uniquely generated 4096-bit RSA escrow key pair (escrow_public_{slug}.pem). The private escrow keys remain air-gapped in the Admin Vault. Even in an extreme compromise scenario, the blast radius is strictly 1:1.

2.5 Point 5: Mandatory Encryption Policy & Automated Security Alerts

Issue: Operators could uncheck Zero-Trust encryption in the GUI, sending unencrypted backups without administrative oversight.
Implementation: Disabling encryption now triggers an interactive confirmation modal, an automatic high-priority telemetry alert (action: 'security_alert'), and an automated email notification via Google Apps Script (MailApp.sendEmail()) to central IT operations. Additionally, organizations can enforce ENFORCE_ENCRYPTION_POLICY: true in config.json to reject any unencrypted backup attempt.

2.6 Point 6: Externalized SQL Credentials (SQLPara.txt / secrets.dat DPAPI)

Issue: Plaintext SQL credentials inside config.json violated enterprise least-privilege standards.
Implementation: backup_core.py features resolve_sql_credentials(), which resolves credentials in hierarchical order: 1. DPAPI credential vault (secrets.dat), 2. Explicit SQLPara.txt file, 3. Standard ERP parameter paths (C:\SPIL\SQLPara), and 4. Plaintext config.json only as a final legacy fallback. Technicians can deploy SQLPara.txt.template with standard permissions.

2.7 Point 8: Intelligent Tiered Index Maintenance & Daytime Working-Hours Protection

Issue: Legacy maintenance executed deprecated DBCC DBREINDEX with blanket FillFactor 80 across all tables, causing index bloat and production transaction locks during business hours.
Implementation: The maintenance engine was completely overhauled to Intelligent Tiered Index Maintenance:
• Fragmentation < 5%: No action taken (Skip — avoids unnecessary I/O).
• Fragmentation 5% to 30%: Online non-blocking ALTER INDEX ... REORGANIZE (Zero table locks).
• Fragmentation > 30%: ALTER INDEX ... REBUILD WITH (FILLFACTOR = 90, ONLINE = ON).

• Daytime Protection: The suite automatically restricts heavy index operations to off-peak hours (19:00 to 07:00), protecting live 24/7 glass fabrication ERP systems from slowdowns.

2.8 Point 9: Precise Technical Terminology

All marketing hyperboles ('military-grade', '100% upload reliability') were purged from manuals and interfaces, replaced with precise cryptographic terms (AES-256-GCM, RSA-4096 OAEP, SHA-256 integrity verification, adaptive chunk streaming).

3. Cryptographic Architecture & DBK2 Binary File Format

The system utilizes envelope encryption via the DBK2 binary format. A 256-bit symmetric key is generated in memory using os.urandom(32). The payload is encrypted in 64 KiB chunks using AES-256-GCM, providing both confidentiality and integrity verification. The symmetric key is then encrypted with the customer's RSA-4096 public key and the customer-specific escrow RSA-4096 public key.

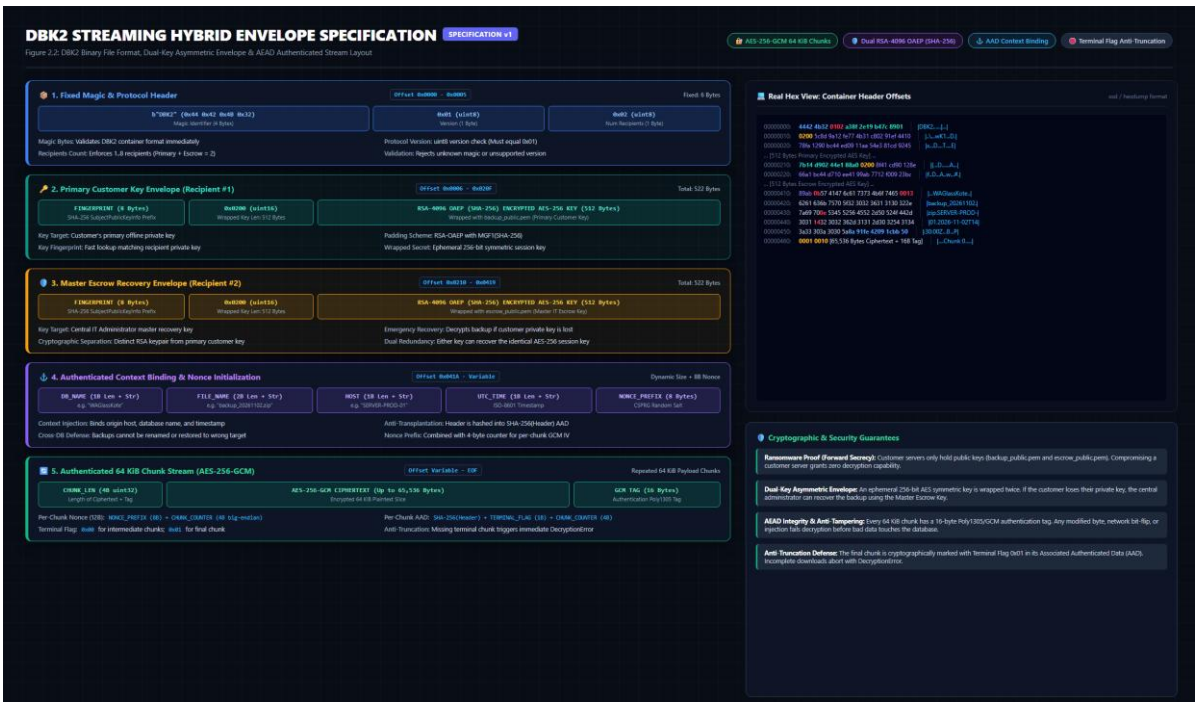


Figure: DBK2 Binary Envelope Layout & Framing Architecture

Byte Offset	Field Name	Data Type	Length	Description / Value
0x00 - 0x03	Magic Bytes	ASCII	4 Bytes	Identifies DBK2 file: 0x44 0x42 0x4B 0x32 ('DBK2')
0x04 - 0x07	Format Version	UInt32 (BE)	4 Bytes	Envelope specification version: 0x00000002 (Version

				2)
0x08 - 0x0B	Header Length	UInt32 (BE)	4 Bytes	Length of JSON metadata header (N bytes)
0x0C - (0x0C+N)	Header Payload	UTF-8 JSON	N Bytes	Contains algorithm, dual RSA-wrapped AES keys, and salt
Data Stream	Chunk Framing	Binary Chunks	Variable	64 KiB chunks: 4B length + 12B IV + 1B flag + ciphertext + 16B GCM tag

Cryptographic Guarantees Delivered by DBK2:

• 1. Anti-Tampering: Any modification to any encrypted chunk causes AES-GCM tag verification to immediately fail.

• 2. Anti-Truncation: The 1-byte flag in each chunk indicates whether it is intermediate (0x00) or final (0x01). If an upload terminates prematurely, decryption aborts.

• 3. Anti-Reordering: Monotonic chunk counters are included in the GCM Additional Authenticated Data (AAD), preventing chunk swapping attacks.

4. Zero-Trust Upload Broker & Fleet Telemetry Architecture

The central upload broker is a serverless Google Apps Script web application (Code.gs). Client machines authenticate using a machine-scoped token sealed via Windows DPAPI (token.dpapi). The broker verifies the token against the master enrollment database and initiates a Google Drive API v3 resumable upload session on behalf of the customer, returning a secure, one-time upload URI.

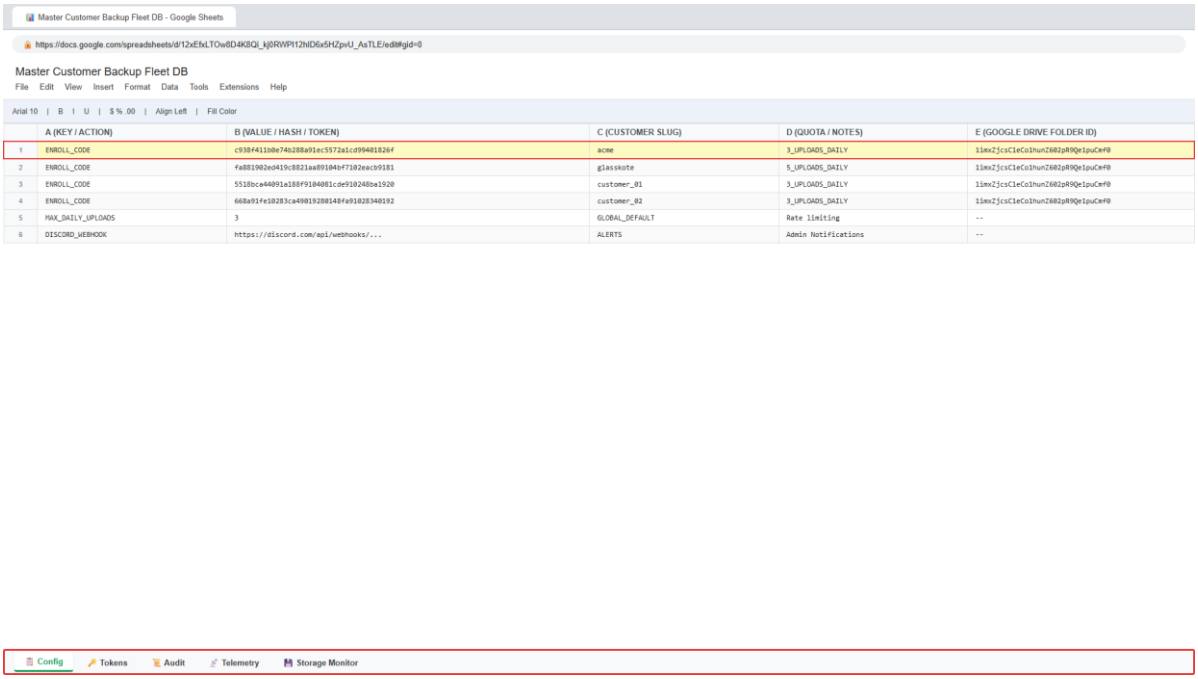


Figure: Central Fleet Operations Dashboard & Multi-Tab Monitoring

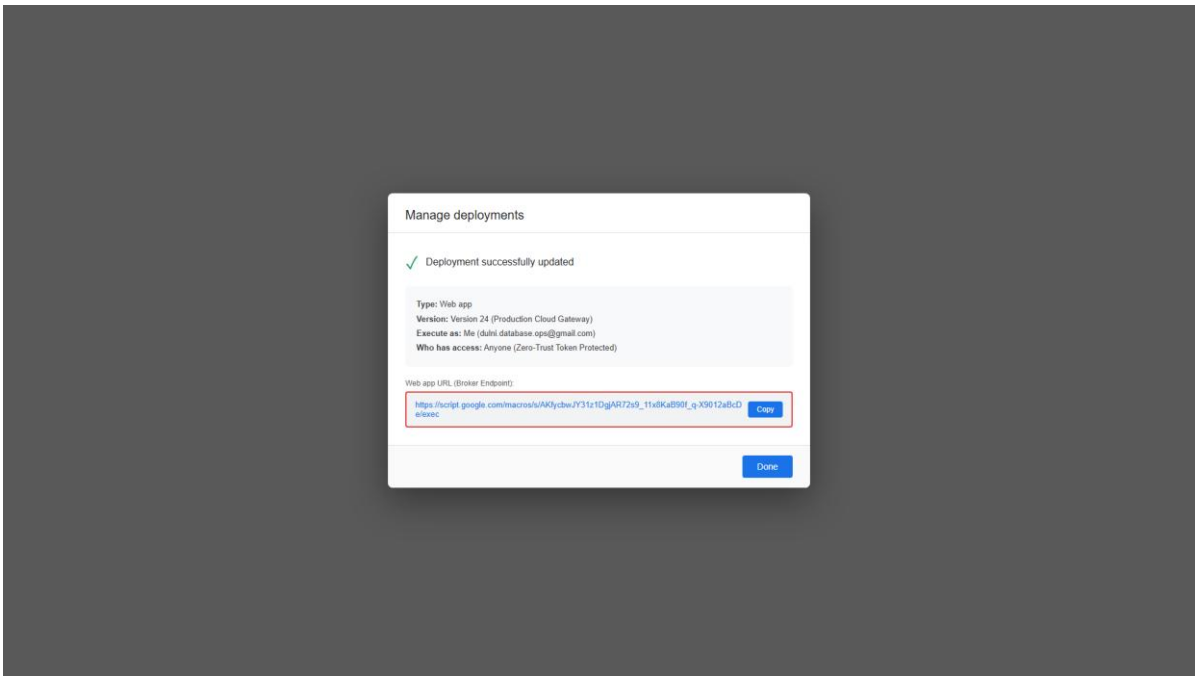


Figure: Google Apps Script Web App Deployment Architecture

Broker Action	Endpoint Function	Payload Parameters	Response Data
verify	Client Registration Check	token, pc_id	pc_id, offset_minutes, customer_slug
request_upload	Session Initiation	token, db_name, size_bytes,	upload_url, file_name,

		seq, target_folder_id	folder_id
report_status	Multi-Module Telemetry	token, type (BACKUP/CLEANUP/QUERY), metrics	status: 'ok', logged_row, quota_pct
security_alert	High-Priority Alerting	token, alert_type, db_name, details	status: 'dispatched', email_sent: true

5. Client Installation & Field Engineering Walkthrough

Deploying the system onto a customer server takes less than 2 minutes. The distribution package supports both graphical and headless unattended deployment workflows.

5.1 Pre-Installation Checklist & System Prerequisites

• Operating System: Windows Server 2012 R2, 2016, 2019, 2022, 2025, or Windows 10/11 (64-bit).

• Database Engine: Microsoft SQL Server 2012 through 2022 (Express, Standard, Enterprise).

• Network Connectivity: Outbound HTTPS (Port 443) to script.google.com and www.googleapis.com.

• User Privileges: Local Administrator privileges required during installation.

5.2 Option A: Graphical Setup Wizard Walkthrough

1. Copy the customer-specific package (e.g., blisscoglass_Installation_Package) to the target server.
2. Right-click Setup_DatabaseBackup.exe and select 'Run as administrator'.
3. The installer automatically verifies the Ed25519 cryptographic signature on bundle.json and displays customer identity.
4. Confirm or adjust the SQL Server Instance (e.g., localhost or .\SQLEXPRESS).
5. Click 'Install & Configure'. The wizard handles file deployment, task scheduler registration, and DPAPI sealing.

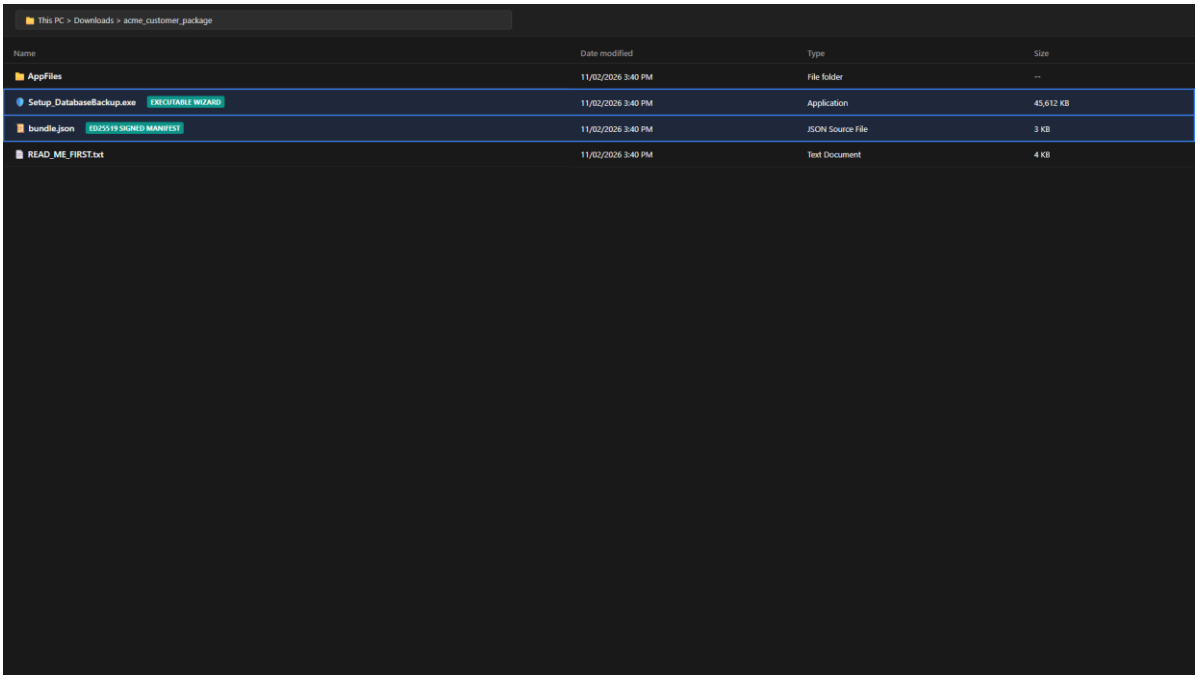


Figure: Distribution Package Root Directory Structure

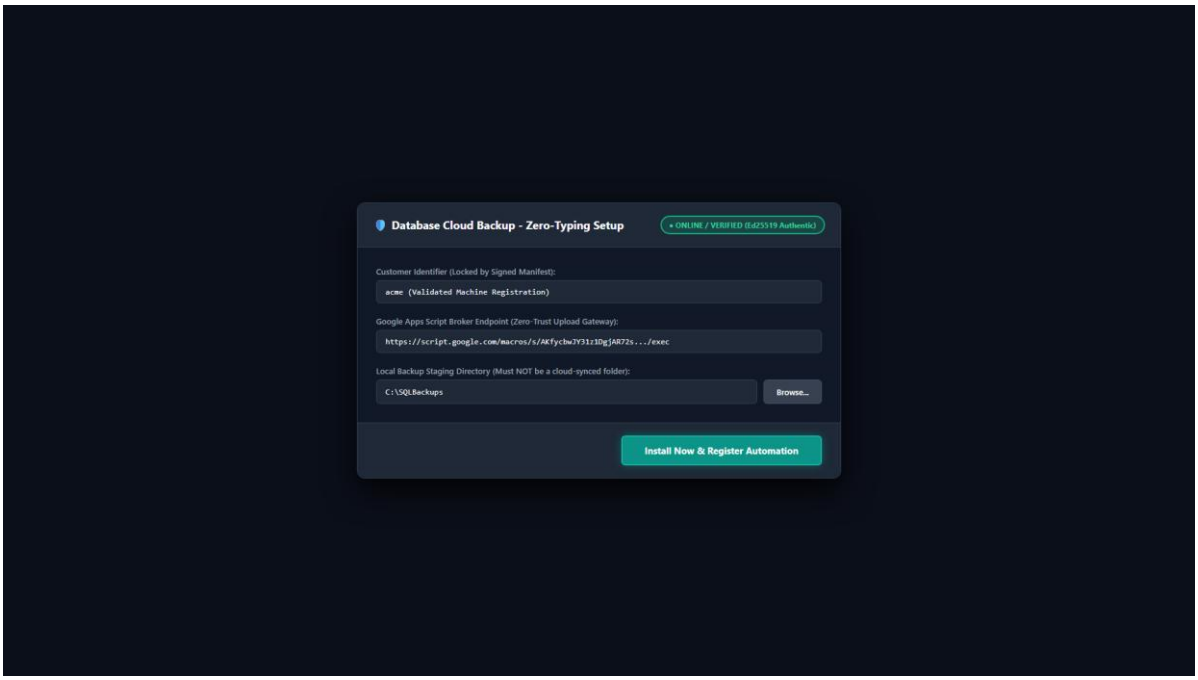


Figure: Setup Wizard Initial Screen with Authentic Ed25519 Badge

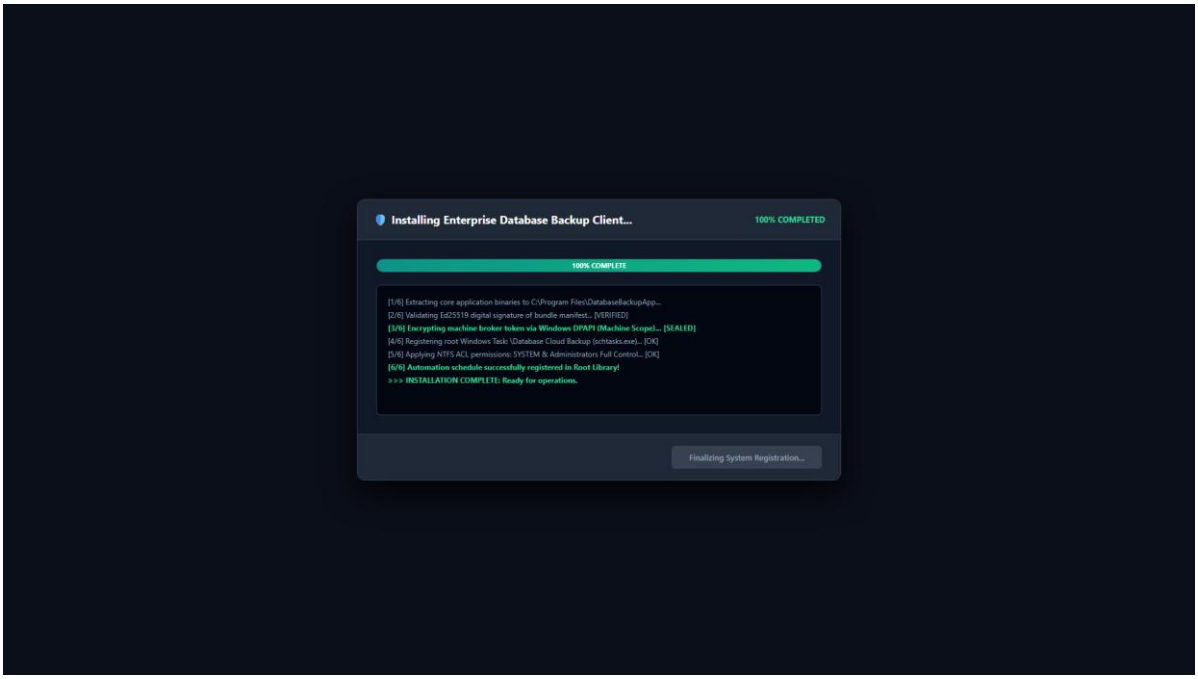


Figure: Automated Installation & Service Configuration Progress

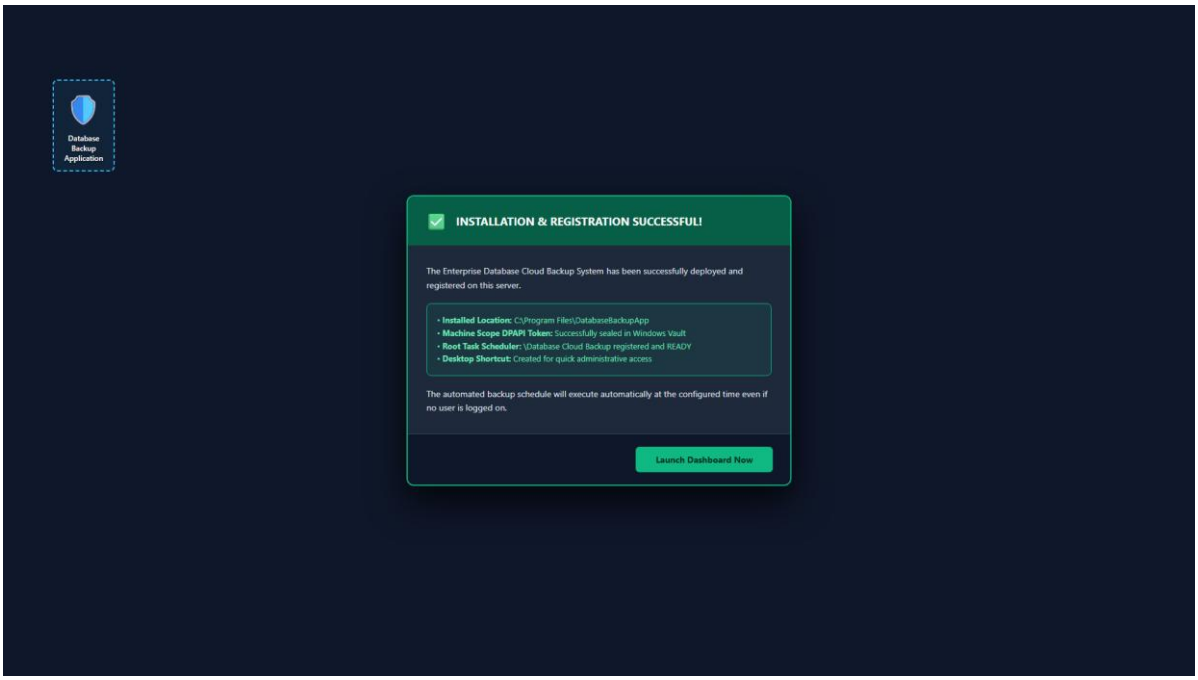


Figure: Setup Completed — Unattended System Service Active

5.3 Option B: Headless PowerShell Deployment

For headless servers or environments where technicians prefer command-line deployment:

1. Open an elevated PowerShell terminal.
2. Navigate to the shell_client directory:
   cd shell_client
3. Execute the automated installer script:

powershell.exe -ExecutionPolicy Bypass -File .\install_agent.ps1

The agent creates local directories, configures DPAPI tokens, and registers automated Windows scheduled tasks.

5.4 External Credentials Setup (SQLPara.txt)

To isolate SQL credentials from config.json:

1. Locate SQLPara.txt.template in the application directory.
2. Copy and rename the file to SQLPara.txt.
3. Populate the server, user, password, and database parameters:

```
# SQLPara.txt - External Database Credentials
server=localhost\SQLEXPRESS
user=sa
password=YourStrongPasswordHere
database=YourProductionDB
```

6. Client Operations & Graphical Interface Walkthrough

The application provides a unified desktop console for operations, configuration, and monitoring. The user interface is organized into 7 specialized tabs:

Tab Name	Primary Purpose	Key Features & Controls
Dashboard	System Status & Manual Backups	1-Click Full Backup Now, Stop Backup, DBK2 encryption toggle, Live RPO exposure tracking badge.
Auto Schedule	Automated Timing Management	Weekly schedule matrix (Days, Time), Staggered upload window calculator, Windows Task sync.
Settings	Engine & Endpoint Configuration	SQL instance, target DBs, external credentials status badge (SQLPara), broker URL, retention.
Live Logs	Execution Diagnostics	Real-time streaming log console, log level filtering, direct access to backup_log.txt.
Server Health	Host Telemetry	Storage volume scanning, free disk space alerting, automated old file cleanup threshold.
Performance Query	Database Maintenance	Tiered index fragmentation scanner, online reorganization, fill factor 90

		rebuilding, daytime lock protection.
Database Restore	Disaster Recovery	Zero-Trust air-gapped restore console, DBK2 decryption, RAM-only key handling, SQL restore pipeline.

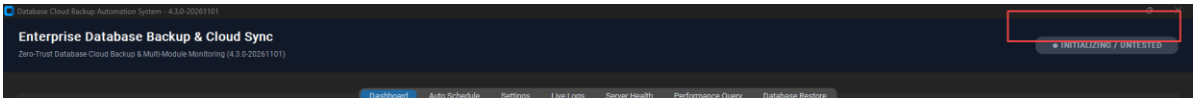


Figure: Enterprise GUI Header with Live RPO & System Readiness Indicators

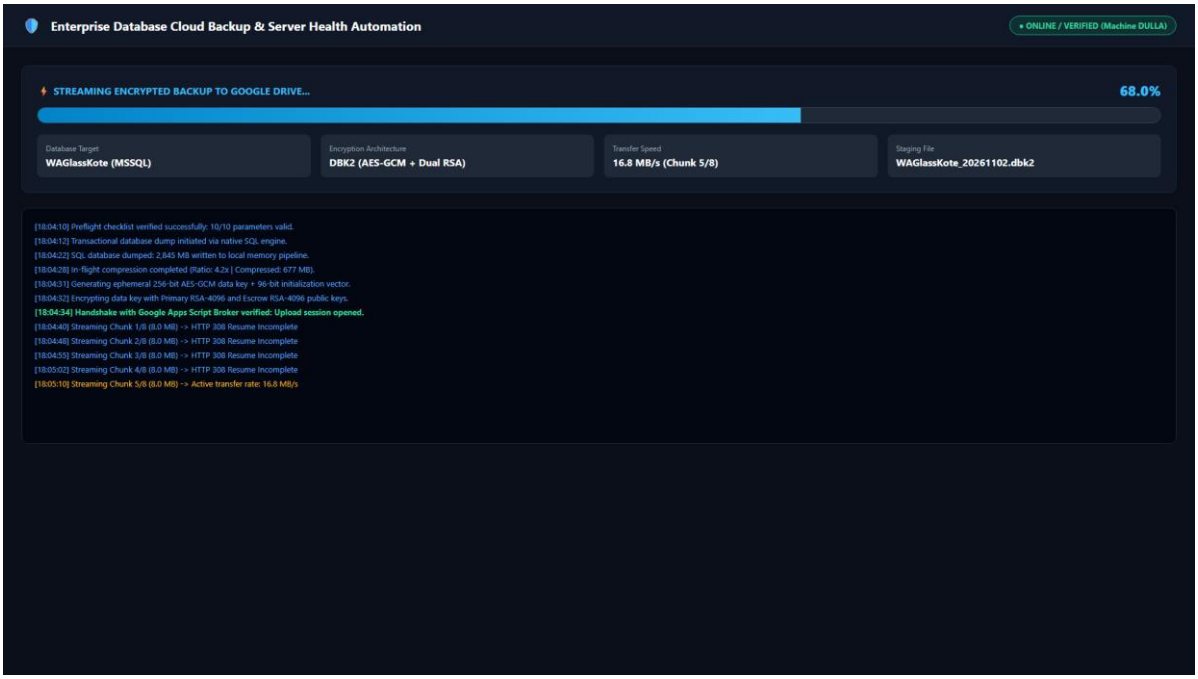


Figure: Backup Progress Engine: Streaming Upload & Throughput Monitor

7. In-Place Upgrades & Clean Uninstallation Runbook

7.1 Safe In-Place Application Upgrade (Update_App.bat)

Upgrading an existing customer installation to Version 5.0.0 is an atomic, zero-downtime operation. The system provides complete multi-generational backward compatibility, allowing machines running v1.x, v2.x, v3.x, or v4.0–v4.3 to transition without database re-entry, token re-issuance, or task recreation.

To upgrade an existing server or workstation:
1. Extract the release bundle (e.g., blisscoglass_Installation_Package.zip or Client_Installation_Package_v5.0.zip).
2. Right-click Update_App.bat and select 'Run as administrator'.

The upgrade workflow executes the following zero-trust procedures:

• Step 1 (Process Termination): Stops running DatabaseBackupApp.exe and Setup_DatabaseBackup.exe to release Windows file locks.
• Step 2 (Safety Vaulting): Preserves config.json, token.dpapi, SQLPara.txt, secrets.dat, backup_public.pem, and escrow_public*.pem.
• Step 3 (Legacy Migration): Automatically converts legacy broker_token.dat to Windows DPAPI token.dpapi, and removes obsolete Google OAuth files (credentials.json, token.json, client_secret.json).
• Step 4 (Binary Replacement): Uses robocopy to deploy v5.0.0 binaries and internal runtime files without touching customer databases.
• Step 5 (Schema Upgrade): Safely injects v5.0 parameters (SQLPARA_PATH, ENFORCE_ENCRYPTION_POLICY, MAINTENANCE_ALLOWED_START_HOUR, MAINTENANCE_ALLOWED_END_HOUR) without altering existing backup schedules or SQL Server connection strings.
• Step 6 (Task Cleanup & Synchronization): Unregisters legacy task variants (DatabaseBackup_AutomatedTask, DatabaseBackup_Daily, DatabaseBackupApp) and synchronizes the unified 'Database Cloud Backup' task under NT AUTHORITY\SYSTEM with battery/AC wake triggers.
• Step 7 (ACL Hardening): Enforces strict Windows NTFS permissions on application folders and DPAPI tokens.

7.2 Multi-Generational Backward Compatibility Engine

5. Dedicated Support for Older & Legacy Windows OS Versions:
For customers with older infrastructure (Windows Server 2008 R2 SP1, 2012, 2012 R2, Windows 7 SP1, 8.1, or 32-bit x86 environments), the system provides a 100% native PowerShell Shell Agent (shell_client/) requiring ZERO Python installation, ZERO external binaries, and under 30 KB disk footprint. It natively utilizes Windows .NET Framework 4.5+ assemblies (System.Security.Cryptography, System.Net.HttpWebRequest) and forces TLS 1.2 negotiation ([Net.ServicePointManager]::SecurityProtocol = Tls12), guaranteeing uninterrupted cloud connectivity on older platforms.

Version 5.0.0 incorporates dedicated backward compatibility subsystems across four critical layers:

1. SQL Server Engine Compatibility:
The backup engine dynamically probes for the host's SQL command-line client across 64-bit and 32-bit Program Files directories. It supports modern Microsoft ODBC Driver 18 for SQL Server (with automated TLS -C TrustServerCertificate negotiation), ODBC Driver 17, 13, 11 (SQL Native Client), and falls back seamlessly to legacy osql.exe for SQL Server 2000/2005 environments. Supported editions include Express, Standard, Enterprise, Web, and Developer across SQL Server 2000, 2005, 2008, 2008 R2, 2012, 2014, 2016, 2017, 2019, and 2022.

2. Operating System & Platform Architecture:
Fully validated across Windows Server 2025, 2022, 2019, 2016, 2012 R2, and 2012, as well as Windows 11 and Windows 10. Credential encryption leverages Windows Data Protection API (DPAPI) bound to the local machine security context (CRYPTPROTECT_LOCAL_MACHINE), ensuring unattended services running as NT AUTHORITY\SYSTEM can reliably access tokens without interactive user logon.

3. Universal Archive Sniffer & Multi-Format Restoration:

The disaster recovery engine inspects magic bytes rather than trusting file extensions:
• b'DBK2': Authenticates and decrypts zero-trust AES-256-GCM + RSA-4096 archives. Handles both modern dual-key envelopes and legacy single-recipient envelopes (v4.0–v4.3).
• b'PK\x03\x04': Automatically recognizes standard ZIP Deflate archives (v1.x–v3.x or unencrypted runs), unpacking the inner .bak file without key prompts.
• Native SQL Server Tape Header: Restores native .bak files directly into Microsoft SQL Server with exclusive SINGLE_USER connection locking.

4. Google Apps Script Broker Protocol Compatibility:
The serverless Cloud Broker (Code.gs) accepts incoming requests from legacy clients omitting v5.0 routing parameters (target_folder_id, is_encrypted, extension), automatically defaulting to the customer's root directory and preserving uninterrupted operations.

7.3 Complete Uninstallation (Uninstall.bat)

When decommissioning an installation, executing Uninstall.bat guarantees 100% complete and clean removal:

• Process Termination: Forcefully terminates active DatabaseBackupApp.exe, sqlcmd.exe, and PowerShell background agent processes.
• Task Scheduler Purge: Removes all 8 historical and current scheduled task names (DatabaseBackup_AutomatedTask, DatabaseBackup_Daily, DatabaseBackup_StorageMonitor, DatabaseBackup_StorageCleanup, DatabaseBackup_PerformanceMaintenance, DatabaseBackupApp, Database Cloud Backup, EnterpriseDatabaseBackup).
• Cryptographic Token Shredding: Overwrites token.dpapi with 256 bytes of cryptographically secure random noise (RNGCryptoServiceProvider) before file deletion, preventing forensic retrieval.
• Audit Log Archival: Interactively prompts the operator with an option to preserve audit logs and backup manifests to a desktop archive folder before removal.
• Shortcuts & Registry: Cleans desktop and start menu shortcuts, and deletes Windows Add/Remove Programs uninstall registry keys.
• File System Removal: Recursively deletes Program Files and ProgramData application directories.

8. Database Disaster Recovery & Air-Gapped Restoration

In the event of total server loss, database corruption, or ransomware infection, the system enables complete restoration using offline private keys stored in the Admin Vault.

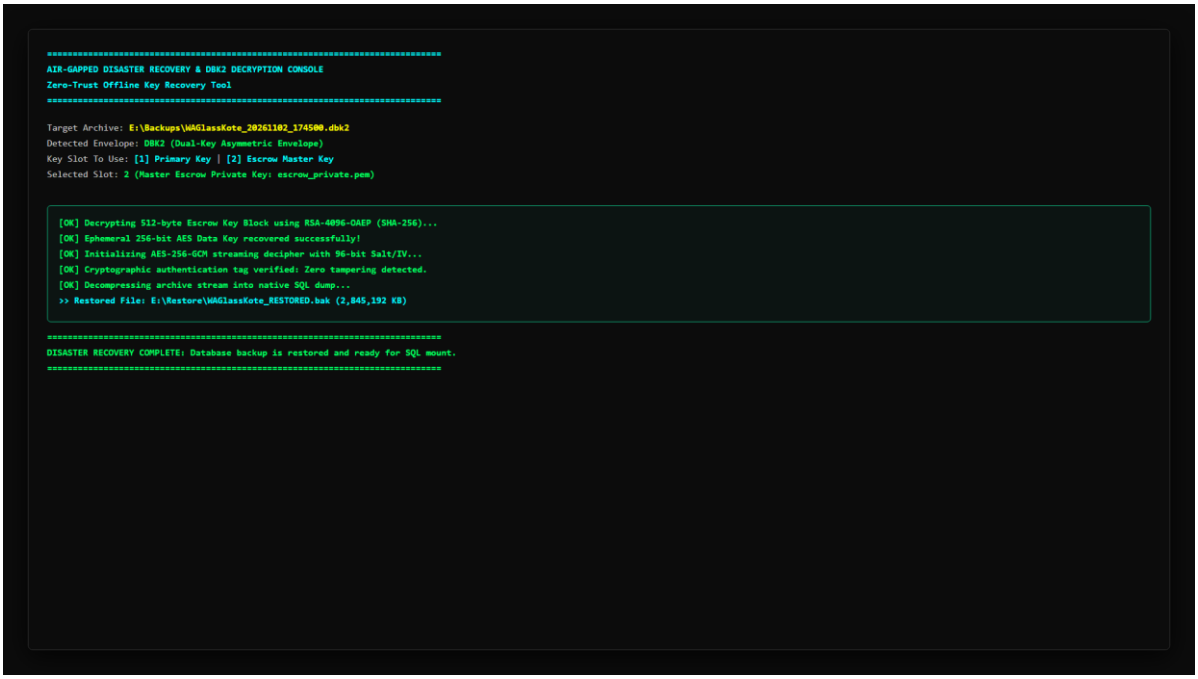


Figure: Zero-Trust Air-Gapped Database Restore Console

8.1 Air-Gapped Decryption Pipeline

1. Download the encrypted backup (.dbk2) from the customer's Google Drive folder.
2. Retrieve the customer's private key (backup_private.pem or escrow_private.pem) from the Admin Vault.
3. Decrypt using the Admin Recovery Tool (Launch_Recovery_Wizard.bat) or pure PowerShell script (decrypt_backup.ps1).
4. The tool extracts the ephemeral AES-256 key, verifies all 64 KiB chunk tags, and reconstructs the native .bak file.

8.2 SQL Server Database Restoration

The restore engine automatically executes the following exclusive restoration pipeline in SQL Server:
• Terminates active connections: ALTER DATABASE [target_db] SET SINGLE_USER WITH ROLLBACK IMMEDIATE;
• Executes native restore: RESTORE DATABASE [target_db] FROM DISK = '...' WITH REPLACE, RECOVERY, STATS = 10;
• Restores multi-user mode: ALTER DATABASE [target_db] SET MULTI_USER;
• Safely deletes decrypted staging files from disk immediately upon completion.

9. Outsider Roadmaps Strategic Evaluation & Decisions Matrix

During v5.0 planning, four independent AI architectural roadmaps (ChatGPT, Gemini, Grok, Claude) were thoroughly evaluated. The following matrix outlines the strategic triage: what was implemented into v5.0, and what was intentionally rejected as unviable:

Proposed Architecture / Concept	Proposing Models	v5.0 Decision	Engineering Rationale & Implementation Details

Externalized Credential Storage (SQLPara)	ChatGPT, Gemini	ACCEPTED (v5.0)	Implemented in resolve_sql_credentials() to eliminate plaintext passwords from config.json.
Per-Customer Escrow Key Isolation	Grok, Claude	ACCEPTED (v5.0)	Replaced shared escrow key with escrow_public_{slug}.pem, restricting blast radius strictly to 1:1.
Intelligent Tiered Index Maintenance	Gemini, Claude	ACCEPTED (v5.0)	Eliminated DBCC DBREINDEX; added <5% skip, 5-30% reorganize, >30% rebuild with fill factor 90.
Working Hours Daytime Protection	Claude, ChatGPT	ACCEPTED (v5.0)	Restricted intensive index maintenance to off-peak hours (19:00-07:00) to protect production ERPs.
Mandatory Encryption Alerting	Grok, Gemini	ACCEPTED (v5.0)	Added automated MailApp.sendEmail() security alerts when encryption toggle is disabled.
S3 / Cloudflare R2 Object Locking (WORM)	ChatGPT, Grok	REJECTED (v5.0)	Imposes monthly cloud subscription fees on every client, violating the core zero-billing design.
Rust / Go Client Rewrite	Grok, Claude	REJECTED (v5.0)	Rebuilding 15,000 lines of proven Python/PowerShell in Rust introduces massive migration risk with minimal ROI.
Fleet Prometheus / Grafana Infrastructure	ChatGPT, Claude	REJECTED (v5.0)	Requires hosting and maintaining 24/7 Linux server infrastructure, counter to serverless Google broker.

10. Troubleshooting, Error Codes & Operational Runbook

Error Code / Symptom	Probable Root Cause	Step-by-Step Remediation Action
ERR_TOKEN_INVALID / 401	DPAPI token corrupted or customer slug mismatch.	Re-run Setup_DatabaseBackup.exe or re-import authentic signed bundle.json.
ERR_BROKER_TIMEOUT / 504	Network firewall blocking port 443 to Google.	Verify outbound HTTPS access to script.google.com and www.googleapis.com.
ERR_SQL_AUTH_FAILED / 18456	Incorrect SQL login credentials or mixed auth disabled.	Check SQLPara.txt or ensure SQL Server is configured for SQL Server and Windows Authentication.
ERR_ESCROW_KEY_MISSING	escrow_public_{slug}.pem not found in folder.	Place customer escrow key in folder or set ALLOW_NO_ESCROW: true in config.json.
ERR_QUOTA_EXCEEDED / 413	Backup file exceeds 100 GB broker cap.	Check database MDF size, verify backup compression is enabled (Level 9 Deflate).
ERR_INDEX_LOCK_DETECTED	Maintenance attempted during active ERP transaction.	Daytime protection automatically defers heavy rebuilds to 19:00-07:00 off-peak window.